THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The ones digit of 127^3 is: 1

(a) 9 **(b)** 1 **(c)** 3 **(d)** 7

2. Which of the following numbers is the greatest? 1

(a) 3.2×10^5 **(b)** 45×10^4 **(c)** 0.5×10^6 **(d)** 480×10^3

3. A number written in base 3 ending in the digit 0 is always divisible by: 1

(a) 4 **(b)** 10 **(c)** 3 **(d)** 2

4. A quadrilateral has equal diagonals that do not bisect each other. It can be an: 1

(a) kite **(b)** isosceles trapezium

(c) rhombus **(d)** parallelogram

5. A number is divisible by both 9 and 11. It must be divisible by: 1

(a) 20 **(b)** 198 **(c)** 99 **(d)** 108

6. The value of $\sqrt{196} + \sqrt{361}$ is: 1

(a) 5 **(b)** 33 **(c)** 266 **(d)** 34

7. If $2^x = 5$, then 2^{x+3} equals: 1

(a) 15 **(b)** 40 **(c)** 13 **(d)** 256

8. If $(25)_b = 17$ in decimal (a two-digit number in base b), then b equals: 1

(a) 7 **(b)** 8 **(c)** 6 **(d)** 5

9. The angles of a quadrilateral are in the ratio 1:3:3:5. The difference between the largest and the smallest angle is: 1

(a) 132° **(b)** 150° **(c)** 120° **(d)** 180°

10. The smallest number which leaves remainder 7 when divided by 15 and 20 is: 1

(a) 127 **(b)** 67 **(c)** 307 **(d)** 60

11. The number of perfect squares between 1000 and 3000 (both excluded) is: 1

(a) 24 **(b)** 23 **(c)** 25 **(d)** 22

12. 3^5 added 3 times equals: 1

(a) 3^8 **(b)** 9^5 **(c)** 3^6 **(d)** 3^{15}

13. **Assertion (A):** $(-1)^{101} = -1$. 1
Reason (R): Any non-zero number raised to the power 0 equals 1.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** The Roman numeral XL represents 40. 1
Reason (R): The Roman symbols V, L and D are never repeated.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The diagonals of a rectangle are equal in length. 1
Reason (R): The diagonals of every parallelogram bisect each other.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the number of non-square numbers between 6^2 and 9^2. 2

17. Find the ones digit of 2^{47}. 2

18. Convert the Roman numeral CXXVII to binary. 2

19. In a kite ABCD, AB = AD, CB = CD, $\angle A = 70^\circ$ and $\angle B = 100^\circ$. Find $\angle C$ and $\angle D$. 2

20. The sum of five consecutive numbers is 45. Find the numbers. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. How many 3-digit numbers are divisible by both 6 and 8? 3

22. A square garden has area 2304 m². Fencing costs ₹35 per metre. Find the total cost of fencing it. 3

23. A sheet of paper 0.2 mm thick is folded in half 8 times (assume this is possible). Find the thickness of the folded paper in mm and in cm. 3

24. A trader has weights of 1, 2, 4, 8, 16, 32 and 64 g. How can she weigh exactly 77 g? Explain using binary numbers. 3

OR

In a parallelogram ABCD with $\angle A = 70^\circ$, the bisectors of $\angle A$ and $\angle B$ meet at P. Find $\angle APB$. 3

25. Show that the six-digit number 358358 is divisible by 7, 11 and 13. 3

OR

A solid cube of side 7 cm is cut into small cubes of side 1 cm. How many small cubes are formed? How many have exactly three painted faces if the big cube was painted all over? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** The foundation stone of a school in Nagpur reads MCMLXXXV, and the plaque on the new building reads XXIV. 5

(i) Convert MCMLXXXV to a Hindu-Arabic number.

(ii) Convert XXIV to a Hindu-Arabic number.

(iii) Write 2024 in Roman numerals.

(iv) The school year is 2024; find the age of the school and write it in Roman numerals.

27. Find the smallest square number that is divisible by each of 4, 9 and 15. 5

OR

A culture has 2 bacteria and the number becomes 4 times every hour. Write the number after 3 hours as an expression with a power, find its value, and say how many times the original number it is. 5

END OF QUESTION PAPER Total: 50 Marks

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